

add_sub_16_bypassed_unsigned	0	0	1	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
32 bit ADDSUB																											
add_sub_32_all_pipelined_unsigned	0	0	1	0	0	1	0	0	0	1	1	0	1	0	0	0	1	0	0	0	0	1	1	1	1	1	1
add_sub_32_all_pipelined_cascaded_unsigned	0	0	1	1	0	1	0	0	0	1	1	0	1	0	0	0	1	0	0	0	0	1	1	1	1	1	1
add_sub_32_all_pipelined_cin_unsigned	0	0	1	1	1	1	0	0	0	1	1	0	1	0	0	0	1	0	0	0	0	1	1	1	1	1	1
add_sub_32_bypassed_unsigned	0	0	1	0	0	1	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
add_sub_32_bypassed_cascaded_unsigned	0	0	1	1	0	1	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
add_sub_32_bypassed_cin_unsigned	0	0	1	1	1	1	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 5: Multiply Add/Subtract Configurations

16 bit Mult-Add/Sub: 8x8 input, 16 bit output Mult-Add/Sub.

32 bit Mult-Add/Sub: 16x16 input, 32 bit output Mult-Add/Sub.

SB_MAC16 Function/ Parameter Settings		B_SIGNED		A_SIGNED		MODE_8x8		BOTADDSUB_CARRYSELECT		BOTADDSUB_UPPERINPUT		BOTADDSUB_LOWERINPUT		BOTOUTPUT_SELECT		TOPADDSUB_CARRYSELECT		TOPADDSUB_UPPERINPUT		TOPADDSUB_LOWERINPUT		TOPOUTPUT_SELECT		PIPELINE_16x16_MULT_REG2		PIPELINE_16x16_MULT_REG1		BOT_8x8_MULT_REG		TOP_8x8_MULT_REG		D_REG		B_REG		A_REG		C_REG	
16 bit Mult-Add/Sub		CBIT[24]	CBIT[23]	CBIT[22]	CBIT[21]	CBIT[20]	CBIT[19]	CBIT[18]	CBIT[17]	CBIT[16]	CBIT[15]	CBIT[14]	CBIT[13]	CBIT[12]	CBIT[11]	CBIT[10]	CBIT[9]	CBIT[8]	CBIT[7]	CBIT[6]	CBIT[5]	CBIT[4]	CBIT[3]	CBIT[2]	CBIT[1]	CBIT[0]													
mult_add_sub_16_all pipelined unsigned		0	0	1	0	0	1	0	1	0	1	0	0	1	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

mult_add_sub_16_intermediate_register_bypassed_unsigned	0	0	1	0	0	1	0	1	0	1	0	0	1	0	1	0	1	0	0	0	0	1	1	1	1
mult_add_sub_16_bypassed_unsigned	0	0	1	0	0	1	0	1	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0
32 bit Mult-Add/Sub																									
mult_add_sub_32_all_pipelined_unsigned	0	0	0	0	0	1	1	0	0	1	1	0	1	1	0	0	1	1	1	1	1	1	1	1	1
mult_add_sub_32_all_pipelined_cascaded_unsigned	0	0	0	1	0	1	1	0	0	1	1	0	1	1	0	0	1	1	1	1	1	1	1	1	1
mult_add_sub_32_all_pipelined_cin_unsigned	0	0	0	1	1	1	1	0	0	1	1	0	1	1	0	0	1	1	1	1	1	1	1	1	1
mult_add_sub_32_intermediate_register_bypassed_unsigned	0	0	0	0	0	1	1	0	0	1	1	0	1	1	0	0	1	1	0	0	0	1	1	1	1
mult_add_sub_32_intermediate_register_bypassed_cascaded_unsigned	0	0	0	1	0	1	1	0	0	1	1	0	1	1	0	0	1	1	0	0	0	1	1	1	1
mult_add_sub_32_intermediate_register_bypassed_cin_unsigned	0	0	0	1	1	1	1	0	0	1	1	0	1	1	0	0	1	1	0	0	0	1	1	1	1
mult_add_sub_32_bypassed_unsigned	0	0	0	0	0	1	1	0	0	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0
mult_add_sub_32_bypassed_cascaded_unsigned	0	0	0	1	0	1	1	0	0	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0
mult_add_sub_32_bypassed_cin_unsigned	0	0	0	1	1	1	1	0	0	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0
mult_add_sub_32_intermediate_register_bypassed_signed	1	1	0	0	0	1	1	0	0	1	1	0	1	1	0	0	1	1	0	0	0	1	1	1	1

Verilog Instantiation

```
SB_MAC16 i_sbmac16
(
    .A(A_i),
    .B(B_i),
    .C(C_i),
    .D(D_i),
    .O(O),
    .CLK(SYSCLK_i),
    .CE(CE_i),
    .IRSTTOP(RST_i),
    .IRSTBOT(RST_i),
    .ORSTTOP(RST_i),
    .ORSTBOT(RST_i),
    .AHOLD(AHOLD_i),
    .BHOLD(BHOLD_i),
    .CHOLD(CHOLD_i),
    .DHOLD(DHOLD_i),
    .OHOLDTOP(HLDTOPOUT_i),
    .OHOLDBOT(HLDBOTOUT_i),
    .OLOADTOP(LOADTOP_i),
    .OLOADBOT(LOADBOT_i),
    .ADDSUBTOP(ADDSUBTOP_i),
    .ADDSUBBOT(ADDSUBBOT_i),
    .CO(CO),
    .CI(CI),
    //MAC cascading ports.
    .ACCUMCI(),
    .ACCUMCO(),
    .SIGNEXTIN(),
    .SIGNEXTOUT()
);

// mult_8x8_all_pipelined_unsigned [24:0] = 001_0000010_0000010_0111_0110
// Read configuration settings [24:0] from left to right while filling the
instance parameters.

defparam i_sbmac16. B_SIGNED                = 1'b0 ;
defparam i_sbmac16. A_SIGNED                = 1'b0 ;
defparam i_sbmac16. MODE_8x8                = 1'b1 ;

defparam i_sbmac16. BOTADDSUB_CARRYSELECT   = 2'b00 ;
defparam i_sbmac16. BOTADDSUB_UPPERINPUT    = 1'b0 ;
defparam i_sbmac16. BOTADDSUB_LOWERINPUT    = 2'b00 ;
defparam i_sbmac16. BOTOUTPUT_SELECT        = 2'b10 ;

defparam i_sbmac16. TOPADDSUB_CARRYSELECT   = 2'b00 ;
defparam i_sbmac16. TOPADDSUB_UPPERINPUT    = 1'b0 ;
defparam i_sbmac16. TOPADDSUB_LOWERINPUT    = 2'b00 ;
defparam i_sbmac16. TOPOUTPUT_SELECT        = 2'b10 ;

defparam i_sbmac16. PIPELINE_16x16_MULT_REG2 = 1'b0 ;
defparam i_sbmac16. PIPELINE_16x16_MULT_REG1 = 1'b1 ;
defparam i_sbmac16. BOT_8x8_MULT_REG        = 1'b1 ;
defparam i_sbmac16. TOP_8x8_MULT_REG        = 1'b1 ;
defparam i_sbmac16. D_REG                    = 1'b0 ;
defparam i_sbmac16. B_REG                    = 1'b1 ;
defparam i_sbmac16. A_REG                    = 1'b1 ;
defparam i_sbmac16. C_REG                    = 1'b0 ;
```

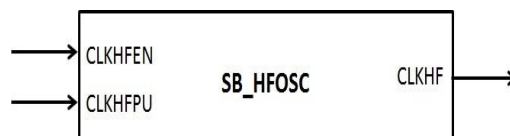
iCE40UL (iCE40 Ultra Lite) Hard Macros

This section describes the following dedicated hard macro primitives available in iCE40UL device.

SB_HFOSC
SB_LFOSC
SB_RGBA_DRV
SB_IR400_DRV
SB_BARCODE_DRV
SB_IR500_DRV
SB_LEDDA_IP
SB_IR_IP
SB_IO_OD
SB_I2C_FIFO

SB_HFOSC

SB_HFOSC primitive generates 48MHz nominal clock frequency within +/- 10% variation, with user programmable divider value of 1, 2, 4, and 8. The HFOSC can drive either the global clock network or fabric routes directly based on the clock network selection.



Ports

SB_HFOSC Ports		
Signal Name	Direction	Description
CLKHFPU	Input	Power up the HFOSC circuit. After power up, oscillator output will be stable after 100us. Active High.
CLKHFEN	Input	Enable the clock output. Enable should be low for the 100us power up period. Active High.
CLKHF	Output	HF Oscillator output.

Parameters

Parameter	Parameter Values.	Description.
CLKHF_DIV	"0b00"	Sets 48MHz HFOSC output.
	"0b01"	Sets 24MHz HFOSC output.
	"0b10"	Sets 12MHz HFOSC output.
	"0b11"	Sets 6MHz HFOSC output

Default Signal Values

The iCEcube2 software assigns the following signal value to unconnected port:

Input CLKHFEN: Logic "0"

Input CLKHFPU: Logic "0"

Clock Network Selection

By default the oscillator use one of the dedicated clock networks in the device to drive the elements. The user may configure the oscillator to use the fabric routes instead of global clock network using the synthesis attribute.

Synthesis Attributes

```
/* synthesis ROUTE_THROUGH_FABRIC = <value> */
```

Value:

0: Use dedicated clock network. Default option.

1: Use fabric routes.

Verilog Instantiation

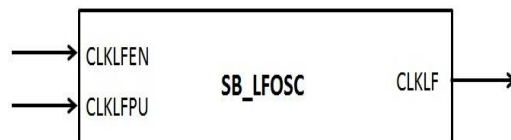
```
SB_HFOSC OSCInst0 (  
  .CLKHFEN(ENCLKHF),  
  .CLKHFPU(CLKHF_POWERUP),  
  .CLKHF(CLKHF)  
) /* synthesis ROUTE_THROUGH_FABRIC= [0|1] */;
```

```
defparam OSCInst0.CLKHF_DIV = "0b00";
```

SB_LFOSC

SB_LFOSC primitive generates 10 KHz nominal clock frequency within +/- 10% variation. There is no divider on the LFOSC.

The LFOSC can drive either the global clock network or fabric routes directly based on the clock network selection



Ports

SB_LFOSC Ports		
Signal Name	Direction	Description
CLKLFPU	Input	Power up the LFOSC circuit. After power up, oscillator output will be stable after 100us. Active High.
CLKLFEN	Input	Enable the clock output. Enable should be low for the 100us power up period. Active High.
CLKLF	Output	LF Oscillator output.

Default Signal Values

The iCEcube2 software assigns the following signal value to unconnected port:

Input CLKLFEN: Logic "0"

Input CLKLFPU: Logic "0"

Clock Network Selection

By default the oscillator use one of the dedicated clock networks in the device to drive the elements. The user may configure the oscillator to use the fabric routes instead of global clock network using the synthesis attribute.

Synthesis Attributes

```
/* synthesis ROUTE_THROUGH_FABRIC = <value> */
```

Value:

0: Use dedicated clock network. Default option.

1: Use fabric routes.

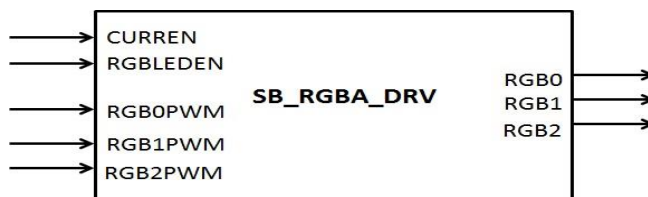
Verilog Instantiation

```
SB_LFOSC OSCInst1 (  
    .CLKLFEN(ENCLKLF),  
    .CLKLFPU(CLKLF_POWERUP),  
    .CLKLF(CLKLF)  
) /* synthesis ROUTE_THROUGH_FABRIC= [0|1] */;
```

SB_RGBA_DRV

SB_RGBA_DRV primitive is the RGB LED drive module which contains 3 dedicated open drain I/O pins for RGB LED outputs. Each of the RGB LED output is bonded out together with an SB_IO_OD primitive to the package pin. User can either use SB_RGBA_DRV primitive or the SB_IO_OD primitive to drive the package pin, but not both.

The primitive allows configuration of each of the 3 RGB LED outputs individually. When the RGBx_CURRENT parameter of RGBx output is set to "0b000000", then SB_IO_OD can be used to drive the package pin.



Ports

SB_RGBA_DRV Ports		
Signal Name	Direction	Description
CURREN	Input	Enable the mixed signal control block to supply reference current to the IR drivers. When it is not enabled (CURREN=0), no current is supplied, and the IR drivers are powered down. Enabling the mixed signal control block takes 100us to reach a stable reference current value.
RGBLEDEN	Input	Enable the SB_RGB_DRV primitive. Active High.
RGB0PWM	Input	Input data to drive RGB0 LED pin. This input is usually driven from the SB_LEDD_IP.
RGB1PWM	Input	Input data to drive RGB1 LED pin. This input is usually

		driven from the SB_LEDD_IP.
RGB2PWM	Input	Input data to drive RGB2 LED pin. This input is usually driven from the SB_LEDD_IP.
RGB0	Output	RGB0 LED output.
RGB1	Output	RGB1 LED output.
RGB2	Output	RGB2 LED output.

Default Signal Values

The iCEcube2 software assigns the following signal value to unconnected port:

Input CURREN : Logic "0"
Input RGBLEDEN : Logic "0"
Input RGB0PWM : Logic "0"
Input RGB1PWM : Logic "0"
Input RGB2PWM : Logic "0"

Parameters

The SB_RGBA_DRV primitive contains the following parameter and their default values:

Parameter CURRENT_MODE = "0b0";

Parameter values:

"0b0" = Full Current Mode (Default).

"0b1" = Half Current Mode.

Parameter RGB0_CURRENT = "0b000000";

Parameter RGB1_CURRENT = "0b000000";

Parameter RGB2_CURRENT = "0b000000";

Parameter values:

"0b000000" = 0mA. // Set this value to use the associated SB_IO_OD instance at RGB
// LED location.

"0b000001" = 4mA for Full Mode; 2mA for Half Mode

"0b000011" = 8mA for Full Mode; 4mA for Half Mode

"0b000111" = 12mA for Full Mode; 6mA for Half Mode.

"0b001111" = 16mA for Full Mode; 8mA for Half Mode

"0b011111" = 20mA for Full Mode; 10mA for Half Mode.

"0b111111" = 24mA for Full Mode; 12mA for Half Mode.

Verilog Instantiation

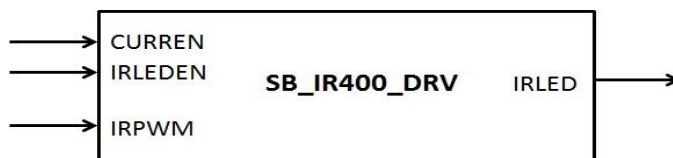
```
SB_RGBA_DRV RGBA_DRIVER (
    .CURREN(ENABLE_CURR),
    .RGBLEDEN(ENABLE_RGBDRV),
    .RGB0PWM(RGB0),
    .RGB1PWM(RGB1),
    .RGB2PWM(RGB2),
    .RGB0(LED0),
    .RGB1(LED1),
    .RGB2(LED2)
);

defparam RGBA_DRIVER.CURRENT_MODE = "0b0";
defparam RGBA_DRIVER.RGB0_CURRENT = "0b111111";
defparam RGBA_DRIVER.RGB1_CURRENT = "0b111111";
defparam RGBA_DRIVER.RGB2_CURRENT = "0b111111";
```


SB_IR400_DRV

SB_IR400_DRV primitive is the IR driver which contains a single dedicated open drain I/O pin for IRLED output. The IRLED output is bonded out together with an SB_IO_OD primitive to the package pin. User can either use SB_IR400_DRV primitive or the SB_IO_OD primitive to drive the package pin, but not both.

When the IR400_CURRENT parameter is set to "0b00000000", then SB_IO_OD can be used to drive the package pin.



Ports

SB_IR400_DRV Ports		
Signal Name	Direction	Description
CURREN	Input	Enable the mixed signal control block to supply reference current to the IR drivers. When it is not enabled (CURREN=0), no current is supplied, and the IR drivers are powered down. Enabling the mixed signal control block takes 100us to reach a stable reference current value.
IRLEDEN	Input	Enable the SB_IR400_DRV primitive. Active High.
IRPWM	Input	PWM Input data to drive IRLED pin.
IRLED	Output	IR LED output.

Default Signal Values

The iCEcube2 software assigns the following signal value to unconnected port:

Input CURREN : Logic "0"
Input IRLEDEN : Logic "0"
Input IRPWM : Logic "0"

Parameter

The SB_IR400_DRV primitive contains the following parameter and their default values:

Parameter CURRENT_MODE = "0b0 ";

Parameter values:

"0b0" = Full Current Mode (Default).

"0b1" = Half Current Mode.

Parameter IR400_CURRENT = "0b00000000";

Parameter Values:

"0b0000000000"; = 0mA. //This is the setting to tristate the IR output to allow it to be used as GPIO (SB_IO_OD)

"0b000000001"; = 50mA for Full Mode; 25mA for Half Mode.

"0b000000011"; = 100mA for Full Mode; 50mA for Half Mode.

"0b000000111"; = 150mA for Full Mode; 75mA for Half Mode.

"0b000011111"; = 200mA for Full Mode; 100mA for Half Mode.

"0b000111111"; = 250mA for Full Mode; 125mA for Half Mode.

"0b001111111"; = 300mA for Full Mode; 150mA for Half Mode.

"0b011111111"; = 350mA for Full Mode; 175mA for Half Mode.

"0b11111111"; = 400mA for Full Mode; 200mA for Half Mode.

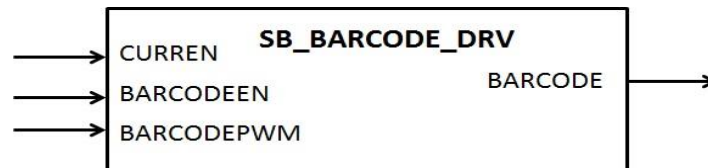
Verilog Instantiation

```
SB_IR400_DRV IRDRVinst (  
    .CURREN(ENABLE_CURRENT),  
    .IRLEDEN(ENABLE_IRDRV),  
    .IRPWM(IR_PWMINPUT),  
    .IRLED(IR_LEDOUT)  
);  
defparam IRDRVinst.CURRENT_MODE = "0b0";  
defparam IRDRVinst.IR400_CURRENT = "0b1111111111";
```

SB_BARCODE_DRV

SB_BARCODE_DRV primitive contains a single dedicated open drain I/O pin for BARCODE output. The BARCODE output is bonded out together with an SB_IO_OD primitive to the package pin. User can either use SB_BARCODE_DRV primitive or the SB_IO_OD primitive to drive the package pin, but not both.

When the BARCODE_CURRENT parameter is set to "0b0000", SB_IO_OD can be used to drive the package pin.



Ports

SB_BARCODE_DRV Ports		
Signal Name	Direction	Description
CURREN	Input	Enable the mixed signal control block to supply reference current to the IR drivers. When it is not enabled (CURREN=0), no current is supplied, and the IR drivers are powered down. Enabling the mixed signal control block takes 100us to reach a stable reference current value.
BARCODEN	Input	Enable the SB_BARCODE_DRV primitive. Active High.
BARCODEPWM	Input	PWM Input data to drive BARCODE pin.
BARCODE	Output	BARCODE output.

Default Signal Values

The iCEcube2 software assigns the following signal value to unconnected port:

Input CURREN : Logic "0"
Input BARCODEEN : Logic "0"
Input BARCODEPWM : Logic "0"